

# Taiki Aiba

[taiba3@gatech.edu](mailto:taiba3@gatech.edu) — 978-201-4007 — [linkedin.com/in/taiba3](https://www.linkedin.com/in/taiba3) — [detoasty3.github.io](https://github.com/detoasty3)

## EDUCATION

- Georgia Institute of Technology**, Atlanta, GA – GPA: 3.92/4.0 2025 - 2027  
M.S. in Mathematics, M.S. in Computer Science
- Georgia Institute of Technology**, Atlanta, GA – GPA: 3.96/4.0 2022 - 2025  
B.S. in Mathematics, B.S. in Computer Science, B.S. in Applied Languages and Intercultural Studies, Minor in Economics  
Studied abroad at **Ritsumeikan Asia Pacific University**, Beppu, Oita, Japan in Summer 2025
- University of Illinois Urbana-Champaign**, Urbana, IL – GPA: 3.9/4.0 2021 - 2022  
B.S. candidate in Mathematics, Minor candidate in Electrical and Computer Engineering; Transferred

## RESEARCH EXPERIENCE

**Quad Packing Research Group**, Remote Spring 2025 - Present  
*Advisors: Timothy Goldberg (Lenoir-Rhyne University) and Lauren Rose (Bard College)*

- Select participants from the 2024 Polymath Jr REU plus other researchers met at JMM 2025.
- Proved an exact expression for the flat ( $r$ -dimensional affine subspaces of a vector space  $V$ ) number, showing that each term  $M(n)$  can be written as the sums and products of certain combinations of powers of 2 in the binary representation of  $n$ .
- Proved quad packing is equivalent to flat packing via an injective mapping, resolving a conjecture in a 2023 MIT Primes paper.
- Resulted in the paper *Quad-packing in the game EvenQuads* (WIP); presented at MAA-SE 2026, contributed to JMM 2026.

**Georgia Tech School of Mathematics**, Atlanta, GA Summer 2025 - Spring 2026  
*Advisor: Ernie Croot*

- Found an upper bound of  $3n^2/4$  for the size of largest induced 4-cycle-free subgraphs in two-dimensional grid graphs by taking  $2 \times 2$  subgrids and orienting them the same way.
- Found an upper bound of  $(n/2)^n$  for the number of largest induced 4-cycle-free subgraphs in two-dimensional grid graphs by considering local  $2 \times 2$  subgrids and identifying a necessary constructive condition.
- Proved upper and lower bounds on the number of such subsets with  $(3/4 - \varepsilon)n^2$  vertices with respect to  $\varepsilon$  by counting based on local  $2k \times 2k$  subgrids and upper bounding binomial coefficients with Stirling bounds.
- Resulted in the paper *4-cycle-free induced subgraphs of grid graphs* ([arXiv](#)); presented at MAA-SE 2026.

**Polymath Jr REU, Williams College**, Remote Summer 2024, 2025, 2026  
*Advisors: Timothy Goldberg (Lenoir-Rhyne University) and Lauren Rose (Bard College)*

- Participant in Summer 2024 and Graduate Research Mentor in Summer 2025 and Summer 2026.
- Mentored 20 undergraduates, introducing them to the research problem and the game itself through resources and weekly meetings.
- Researched the quad number  $q(n)$  in the card game *Quads*; studied affine geometry and *Quads* cards as bit vectors in  $\mathbb{Z}_2^6$ .
- Wrote C++ code to brute-force results for  $n \in [4, 16]$  and  $d = \lceil \log_2 n \rceil$ ; used these results to conjecture that  $d$  does not matter and  $q(2^n + 1) = q(2^n)$  for all  $n$  and to prove an upper bound expression for  $q(n)$  and attainability for  $n$  a power of two.
- Resulted in the unpublished paper *Quad Packing* ([paper](#)); presented at JMM 2025 and MAA-SE 2025.

**Georgia Tech School of Mathematics**, Atlanta, GA Fall 2024 - Spring 2025  
*Advisor: Ernie Croot*

- Researched maximal induced forests in multi-dimensional grid graphs and found and proved upper and lower bounds on  $d$  dimensions with the hand-shaking lemma and considering sums of coordinates mod  $2d + 1$ .
- Found a zig-zag construction giving a tight upper bound of  $(2n^2 + O(n))/3$  for two-dimensional graphs.
- Worked on the paper *Maximal induced forests in grid graphs*, but scrapped it after discovering our results were already found in the paper *New bounds on the size of the minimum feedback vertex set in meshes and butterflies* by Caragiannis *et al.* (2002).

**Georgia Tech School of Mathematics**, Atlanta, GA Fall 2023 - Spring 2024  
*Advisor: Ernie Croot*

- Studied grid-counting problems (e.g., rook problem) and searched for bijections to spanning trees (à la Temperley's bijection) in order to apply the Kirchhoff Matrix-Tree Theorem.
- Explored the Prouhet-Tarry-Escott problem and attempted to find results using contour integration and algebraic bounding.
- Explored the Kirchhoff Matrix-Tree Theorem and researched domino tiling enumerations on a grid to attempt to strengthen existing bounds (used this problem to transition to researching grid graphs).

**Georgia Tech School of Modern Languages**, Atlanta, GA Summer 2024  
*Advisor: Kyoko Masuda*

- Researched frequency of female versus male pronouns in the manga *Sazae-san* and drew conclusions based on different age groups.
- Submitted collaborative research paper in Japanese to the SGSJ in 2024; presented at SGSJ 2024.

**Georgia Tech School of Electrical and Computer Engineering**, Atlanta, GA Fall 2023  
*Advisor: Aaron Lanterman*

- Translated documents of the PC-FX game console from Japanese to English, compiled in a GitHub repository for general usability.
- Analyzed the C Compiler of the GMAKER Starter Kit (processing flow/registers), allowing user-made software to run.

READING EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Georgia Tech School of Mathematics, Atlanta, GA                                                                                                                                                                                                                                                                                                                                                                                              | Spring 2025             |
| Advisor: Xingxing Yu                                                                                                                                                                                                                                                                                                                                                                                                                         |                         |
| <ul style="list-style-type: none"><li>Wrote the unpublished paper <i>On the Turán Number of the Blow-Up of the Hexagon</i> (paper), surveying Janzer <i>et al.</i> (2022).</li><li>Gave a one-hour whiteboard presentation to a class of ~10 students; collaborated with a Ph.D. student and a post-doctorate.</li></ul>                                                                                                                     |                         |
| Independent Reading, Atlanta, GA                                                                                                                                                                                                                                                                                                                                                                                                             | Summer 2023             |
| <ul style="list-style-type: none"><li>Read about Burnside’s lemma, including applications to classic problems and competition problems (AMC/AIME).</li><li>Wrote the unpublished paper <i>Burnside’s lemma and applications in competition problems</i> (paper); presented at MAA-SE 2026.</li></ul>                                                                                                                                         |                         |
| Georgia Tech Directed Reading Program, Atlanta, GA                                                                                                                                                                                                                                                                                                                                                                                           | Fall 2022 - Spring 2025 |
| <ul style="list-style-type: none"><li>Read and presented on generating functions in combinatorics and algebra in Fall 2022, combinatorial problem solving techniques in Spring 2023 (slides), the Union-Closed Sets Conjecture in Fall 2023 (slides), Sudoku and graph theory in Spring 2024 (slides), quantitative finance in Fall 2024, and data structures/algorithms, focusing on the Gomory-Hu Tree, in Spring 2025 (slides).</li></ul> |                         |

PUBLICATIONS

|                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|
| 1. T. Aiba and E. Croot, <i>4-cycle-free induced subgraphs of grid graphs</i> , arXiv:2604.08397, submitted to <i>Ars Combinatoria</i> . |
|------------------------------------------------------------------------------------------------------------------------------------------|

CONFERENCES

|                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. <i>Burnside’s lemma and applications in competition problems</i> , MAA Southeastern Sectional Meeting, University of North Alabama, Florence, AL, Contributed Papers Session #6, March 2026. (slides) |
| 2. <i>Quad-packing in the game EvenQuads</i> , MAA Southeastern Sectional Meeting, University of North Alabama, Florence, AL, Recreational Math Session #2, March 2026. (slides)                         |
| 3. <i>4-cycle-free induced subgraphs of grid graphs</i> , MAA Southeastern Sectional Meeting, University of North Alabama, Florence, AL, Contributed Papers Session #1, March 2026. (slides)             |
| 4. <i>Quad Packing</i> , MAA Southeastern Sectional Meeting, High Point University, High Point, NC, Undergraduate Paper Session II-4, March 2025. (slides)                                               |
| 5. <i>Quad Packing</i> , Joint Mathematics Meetings, Seattle Convention Center, Seattle, WA, AMS Special Session on Polymath Jr REU Student Research Session, I, January 2025. (slides)                  |
| 6. 「サザエさん」の女性登場人物が用いるジェンダー表現の考察, The Society for Gender Studies in Japanese Conference, Tokyo International University, Toshima, Tokyo, Japan, July 2024.                                                |

AWARDS

|                                                                                                                                  |              |
|----------------------------------------------------------------------------------------------------------------------------------|--------------|
| Polymath Jr REU Mentor Stipend (\$2500)                                                                                          | May 2026     |
| Awarded to one graduate mentor per project. In Summer 2026, I was the paid mentor for the Games and Finite Geometry project.     |              |
| Outstanding Graduating Mathematics Major (\$100)                                                                                 | April 2025   |
| Awarded to select outstanding undergraduate senior mathematics majors annually. In 2025, I was one of five award recipients.     |              |
| Outstanding Senior in Japanese                                                                                                   | April 2025   |
| Awarded to one outstanding undergraduate senior annually in the Japanese department.                                             |              |
| University of Illinois Freshman Math Contest Third Place                                                                         | October 2021 |
| Ranked third out of 22 first-years in a five-question proof-based math contest across the three University of Illinois campuses. |              |

TEACHING

|                                                                                                                                                                                                                                                                                                                                                                                 |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Georgia Institute of Technology, Atlanta, GA                                                                                                                                                                                                                                                                                                                                    | Spring 2023 - Spring 2026 |
| <ul style="list-style-type: none"><li>CS 3510: Design and Analysis of Algorithms (Teaching Assistant: Fall 2024, Spring 2025, Fall 2025, Spring 2026)</li><li>CS 2050: Introduction to Discrete Mathematics for Computer Science (Teaching Assistant: Summer 2023, Fall 2023, Spring 2024)</li><li>MATH 2551: Multivariable Calculus (Lecture Assistant: Spring 2023)</li></ul> |                           |
| University of Illinois Urbana-Champaign, Urbana, IL                                                                                                                                                                                                                                                                                                                             | Spring 2022               |
| <ul style="list-style-type: none"><li>CS 124: Introduction to Computer Science I (Course Assistant: Spring 2022)</li></ul>                                                                                                                                                                                                                                                      |                           |

WORK EXPERIENCE

|                                                              |                             |
|--------------------------------------------------------------|-----------------------------|
| IBM, Software Engineer Intern, San Jose, CA                  | August 2026 - December 2026 |
| Capital One, Software Engineer Intern, McLean, VA            | June 2026 - August 2026     |
| Amazon, Software Engineer Intern, Boston, MA                 | June 2024 - August 2024     |
| Art of Problem Solving, Math Teaching Assistant, Waltham, MA | July 2022 - Present         |
| Math League, LLC, Curriculum Specialist, Remote              | July 2021 - Present         |

CERTIFICATIONS

|                                                            |                             |
|------------------------------------------------------------|-----------------------------|
| Akuna Capital, Options 201                                 | September 2024              |
| MIT Professional Learning, No Code AI and Machine Learning | August 2023 - December 2023 |

## SERVICE

---

- Mathematical Association of America**, AMC 10/12 Editorial Board Member, Remote September 2024 - Present
- Propose problems for the AMC 10/12 and attend meetings to discuss proposals and determine which problems are suitable.
- MATHCOUNTS Foundation**, Question Writing Committee Member, Remote February 2025 - Present
- Propose problems for the MATHCOUNTS Competition Series and attend in-person meetings to review and edit the problems.
  - Volunteer at the annual MATHCOUNTS National competition held in-person.
- Mustang Math**, Problem Writing Team Member, Remote July 2024 - Present
- Assist with hosting math contests primarily aimed at middle school students; propose, edit, and review problems for the contests.
- De Mathematics Competitions**, Founder, Director, and Tutor, Remote September 2020 - Present
- Write and direct mock American Mathematics Competitions contests on the website Art of Problem Solving.
  - Tutor a third-grade student for the AMC 8, a contest primarily aimed at middle school students.
- Georgia Tech Undergraduate Mathematics Advisory Committee**, President, Atlanta, GA Fall 2023 - Spring 2025
- Hosted 2-4 social events (e.g., puzzle tournaments, graduate student dinners, parties) for math majors every semester.
  - Led and handled event logistics, redesigned and edited the UMAC website with photos, and designed flyers for all events.
- Georgia Tech High School Math Day**, Student Volunteer, Atlanta, GA Spring 2023, 2024, 2025, 2026, 2027
- Served as a problem writer and a volunteer on the day of for a math contest taken by 300+ high school students annually.
  - Wrote problems for the contest, proctored, collected papers, and graded submissions.
- Georgia Tech Japan Student Association**, Executive Board Officer, Atlanta, GA Fall 2023 - Spring 2027
- Led weekly conversation practices to enrich Japanese learners' conversation skills.
  - Assisted in hosting occasional special events related to the Japanese language and culture.

## SKILLS

---

**Programming:** C/C++, Python 3, Java, Go, Rust, JavaScript/TypeScript, SQL, R, Verilog/SystemVerilog

**Libraries:** NumPy, pandas, PyTorch, TensorFlow, scikit-learn, SimPy, Matplotlib

**Technologies:** React, Node.js, Spring Boot, Firebase, Docker, GCP, MongoDB, Git, Bash, Linux, Jira

**Languages:** English (native), Japanese (advanced), Spanish (beginner)

**Interests:** Set (winner of JMM 2025 Set Championships), running, drawing, blogging, electric guitar (9 years), piano (9 years), writing math problems, competitive programming